General Bite #11: Directional Antennas (Yagi And Stacks)

Driven element, reflector, director; gain, bandwidth, front-to-back, main lobe; dBd vs dBi; matching (gamma and beta).

Group G9C · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Yagi Element Lengths

A Yagi has one **driven element** fed by the coax, plus parasitic elements (reflectors behind, directors in front).

Driven element length: Approximately 1/2 wavelength (G9C02).
Reflector and director compared to driven element: The reflector is longer, and the director is shorter (G9C03). The slightly-detuned parasitic elements re-radiate with a phase shift that reinforces forward radiation and cancels rearward radiation.

Bandwidth And Gain

Question	Answer	Q
How to increase Yagi bandwidth	**Larger-diameter elements** (G9C01). Fatter elements have lower Q.	G9C01
Primary effect of longer boom + more directors	**Gain increases** (G9C05). More directors and more boom length = higher forward gain.	G9C05
What can be adjusted to optimize forward gain, F/B, SWR bandwidth?	Boom length + number of elements + spacing – **all of the above** (G9C10).	G9C10

dBi vs dBd

Gain in dBi compared to dBd for the same antenna: Gain in dBi is 2.15 dB higher (G9C04).
dBd = decibels over a half-wave dipole. dBi = decibels over an isotropic radiator. A half-wave dipole has 2.15 dB gain over isotropic, so for the same antenna the dBi value is always 2.15 dB larger than the dBd value.

Pattern Vocabulary

Term	Meaning	Q
Main lobe	**The direction of maximum radiated field strength from the antenna** (G9C08).	G9C08
Front-to-back ratio	**The power radiated in the major lobe compared to that in the opposite direction** (G9C07). Higher = more rejection from behind.	G9C07

Stacked Yagis

Gain of two 3-element Yagis stacked vertically 1/2 λ apart vs a single 3-element Yagi: Approximately 3 dB higher (G9C09). Doubling the antenna count for the same power produces +3 dB gain when correctly phased and spaced.

Matching The Yagi To 50 Ω Coax

Match	How it works	Q
Beta (hairpin) match	**A shorted transmission line stub placed at the feed point of a Yagi antenna to provide impedance matching** (G9C11). Adds inductance to cancel the driven element's capacitive reactance.	G9C11
Gamma match	**Does not require the driven element to be insulated from the boom** (G9C12). Coax taps to the driven element through an offset gamma rod; the element does not need to be insulated from the boom.	G9C12

Mental Cheat Sheet

- Driven element ≈ **1/2 λ**.
- Reflector **longer**; director **shorter**.
- Larger element diameter → **wider bandwidth**.
- Longer boom + more directors → **higher gain**.
- Boom + element count + spacing all tunable for gain/F-B/SWR.
- dBi = dBd + **2.15** (for the same antenna).
- Main lobe = direction of max field strength.
- F/B ratio = main / rear power ratio.
- Two stacked 3-el Yagis 1/2 λ apart: **+3 dB**.
- Beta/hairpin match = **shorted stub** at feed point.
- Gamma match: driven element **stays bonded to boom**.

Common Traps

- Reflector is **longer** than driven; director is **shorter**. Easy to swap.
- Bandwidth grows with **larger** element diameter, not "closer spacing" or "loading coils."
- dBi is **higher** than dBd for the same antenna; the trap says "dBi is lower."
- Stacking two Yagis gives **3 dB**, not 6 dB (that's quadrupling).
- Beta/hairpin = **shorted stub**. Gamma = different system that keeps element bonded.
- F/B is **main vs rear**, not "number of directors vs reflectors."

Takeaway: A Yagi has a 1/2 λ driven element, a longer reflector, and shorter directors. More boom and more directors → more gain. Element diameter sets bandwidth. dBi is always 2.15 dB higher than dBd for the same antenna. Stacking two identical Yagis 1/2 λ apart adds 3 dB. Match with a beta/hairpin (shorted stub) or gamma (coax center conductor tapped to driven element at an offset point via the gamma rod).

Bite 11 Question Bank (G9C)

G9C01. Which of the following would increase the bandwidth of a Yagi antenna?
A. Larger-diameter elements.
B. Closer element spacing.
C. Loading coils in series with the element.
D. Tapered-diameter elements.

G9C02. What is the approximate length of the driven element of a Yagi antenna?
A. 1/4 wavelength.
B. 1/2 wavelength.
C. 3/4 wavelength.
D. 1 wavelength.

G9C03. How do the lengths of a three-element Yagi reflector and director compare to that of the driven element?
A. The reflector is longer, and the director is shorter.
B. The reflector is shorter, and the director is longer.
C. They are all the same length.
D. Relative length depends on the frequency of operation.

G9C04. How does antenna gain in dBi compare to gain stated in dBd for the same antenna?
A. Gain in dBi is 2.15 dB lower.
B. Gain in dBi is 2.15 dB higher.
C. Gain in dBd is 1.25 dBd lower.
D. Gain in dBd is 1.25 dBd higher.

G9C05. What is the primary effect of increasing boom length and adding directors to a Yagi antenna?
A. Gain increases.
B. Beamwidth increases.
C. Front-to-back ratio decreases.
D. Resonant frequency is lower.

G9C07. What does “front-to-back ratio” mean in reference to a Yagi antenna?
A. The number of directors versus the number of reflectors.
B. The relative position of the driven element with respect to the reflectors and directors.
C. The power radiated in the major lobe compared to that in the opposite direction.
D. The ratio of forward gain to dipole gain.

G9C08. What is meant by the “main lobe” of a directive antenna?
A. The magnitude of the maximum vertical angle of radiation.
B. The point of maximum current in a radiating antenna element.
C. The maximum voltage standing wave point on a radiating element.
D. The direction of maximum radiated field strength from the antenna.

G9C09. In free space, how does the gain of two three-element, horizontally polarized Yagi antennas spaced vertically 1/2 wavelength apart typically compare to the gain of a single three-element Yagi?
A. Approximately 1.5 dB higher.
B. Approximately 3 dB higher.
C. Approximately 6 dB higher.
D. Approximately 9 dB higher.

G9C10. Which of the following can be adjusted to optimize forward gain, front-to-back ratio, or SWR bandwidth of a Yagi antenna?
A. The physical length of the boom.
B. The number of elements on the boom.
C. The spacing of each element along the boom.
D. All these choices are correct.

G9C11. What is a beta or hairpin match?
A. A shorted transmission line stub placed at the feed point of a Yagi antenna to provide impedance matching.
B. A 1/4 wavelength section of 75-ohm coax in series with the feed point of a Yagi to provide impedance matching.
C. A series capacitor selected to cancel the inductive reactance of a folded dipole antenna.
D. A section of 300-ohm twin-lead transmission line used to match a folded dipole antenna.

G9C12. Which of the following is a characteristic of using a gamma match with a Yagi antenna?
A. It does not require the driven element to be insulated from the boom.
B. It does not require any inductors or capacitors.
C. It is useful for matching multiband antennas.
D. All these choices are correct.

Answer Key

G9C01: A	G9C02: B	G9C03: A	G9C04: B
G9C05: A	G9C07: C	G9C08: D	G9C09: B
G9C10: D	G9C11: A	G9C12: A	